

1. You will find these problems addressed via the Dashboard feature, Monthly Reports Summary and Waste collection logging feature. The Dashboard alone has solved two problems: The inability to measure campaign impact and lack of location-level comparison by providing a map tracker and allowing the school to view real time data on the dashboard. Along with the Monthly Report Summaries allowing them to determine whether a campaign was successful or not. As for the unstructured data collection, the Waste collection logging feature allowed the staff a structured and more convenient way to log in their reports.

2.

```
SELECT c.volume_kg, l.location_name, l.location_type
FROM Collection_Event c
INNER JOIN Location l
GROUP BY l.location_id
HAVING COUNT (c.event_id) < 1
ON c.location_id = l.location_id
ORDER BY SUM(volume_kg) DESC;
```

3.

```
SELECT i.status, s.staff_name, i.report_type, i.location_id, i.report_date
FROM Issue_Report i
INNER JOIN Staff s
ON i.staff_id = s.staff_id
WHERE status = 'Pending' AND 'In Progress'
ORDER BY i.report_date DESC;
```

4.

```
CREATE TABLE Student (
student_id INT AUTOINCREMENT PRIMARY KEY,
student_name VARCHAR(100) NOTNULL,
student_email VARCHAR(100) NOTNULL,
role_id INT NOTNULL,
FOREIGN KEY (role_id) REFERENCES Role(role_id) ON UPDATE CASCADE
);
```

```
CREATE TABLE Student_Issue_Report(
report_id INT AUTOINCREMENT PRIMARY KEY,
student_id INT NOTNULL,
Location_id INT NOTNULL,
report_type VARCHAR(100) NOTNULL,
description TEXT NOTNULL,
report_date DATETIME DEFAULT CURRENT_TIMESTAMP,
status ENUM('Pending', 'Resolved', 'In Progress') DEFAULT 'Pending',
```

```
FOREIGN KEY (student_id) REFERENCES Student(student_id),  
FOREIGN KEY (location_id) REFERENCES Location(location_id)  
);
```

5. Simply put the data missing would be the campaign data itself, more specifically the record of campaigns that have been initiated within the school. Making it so that yes the school MAY have the statistics of improvements but they cannot properly or accurately determine what campaign took place during that time slot.

```
CREATE TABLE Campaigns (  
Campaign_id INT AUTOINCREMENT PRIMARY KEY,  
location_id INT NOTNULL,  
Campaign_desc TEXT NOTNULL,  
Campaign_title VARCHAR (100),  
event_date DATETIME,  
Remarks TEXT;  
FOREIGN key (location_id) REFERENCES Location(location_id)  
);
```

It connects to my collection event table via the location_id, event_date, and the remarks. The idea is that school staff and administrators can make comparisons based on the remarks before and after the event as well as comparing the locations before and after the event. They can also make a good comparison of before and after between the date of the events via the event_date.